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PHARMACOEPIDEMIOLOGY AND PRESCRIPTION

Non-steroidal anti-inflammatory drug-associated acute kidney injury: does short-term NSAID use pose a risk in hospitalized patients?

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Abstract

Purpose While renal risk associated with short-term use of non-steroidal anti-inflammatory drugs (NSAID) has been anecdotally documented, no conclusive evidence is available on the renal safety, especially among hospitalized patients with reduced renal function. This study is to evaluate the risk of acute kidney injury (AKI) associated with NSAID use in hospital.

Methods A retrospective matched cohort study utilizing electronic health records from two large academic tertiary-care hospitals was conducted. We defined AKI based on changes in SCr according to published AKI criteria. The hospital acquired AKI risk associated with inpatient NSAID use was assessed using a time-dependent Cox proportional hazard regression in pooled cohort as well as two sub cohorts stratified by baseline renal function.

Results A total of 18,794 admissions were included in the final cohort. Of 9397 admissions exposed to NSAIDs, 7914 and 1483 admissions were in the "without" and "with baseline renal impairment" cohort, with the same number of matching non-exposed admissions in each of the stratified cohort. The AKI incidences were 6 and 22 events per 1000 patient-days in "without" and "with preexisting renal impairment" cohort, respectively. The adjusted analyses suggested that NSAID use increased AKI risk in patients with preexisting renal impairment (hazard ratio [HR]: 1.38 [1.04–1.83]) but not in the patients without preexisting renal impairment (HR: 0.83 [95% CIs: 0.63–1.08]) or in the pooled cohort (HR: 1.01 [95% CIs: 0.83–1.24]).

Conclusion Our findings suggested that NSAID use is associated with an increased risk of AKI in the hospitalized patients with preexisting renal impairment but the association is not statistically significant in those who have preserved renal function. Further randomized controlled trials are needed to validate these observational findings.

Keywords Non-steroidal anti-inflammatory drugs · Nephrotoxicity · Drug safety · Outcomes research/analysis

Introduction

Chronic use of NSAIDs has been associated with increased risk of acute kidney injury (AKI) even in patients with preserved renal function [1–3]. The pathogenesis of NSAID-induced AKI includes reduced renal plasma flow, intraglomerular autoregulation interference, and renal toxicity, raising the risk of decreased glomerular filtration rate (GFR), acute tubular necrosis, or intestinal nephritis [4]. Renal risk associated with short-term use (i.e., less than 7 consecutive days) of several drugs in this class such as ibuprofen and celecoxib has been anecdotally documented [5, 6]. However, no conclusive evidence is available on the renal safety of their short-term use among patients with reduced renal function.

Hospitalized patients are particularly vulnerable to AKI development owing to concurrent renal insults during acute care episodes. For example, advanced age, infections (e.g., sepsis), hypotension, abnormal electrolyte level, renal

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ischemia secondary to surgery or other procedures, and exposure to multiple drugs are common in inpatient settings and known to increase AKI risk [7]. In addition to contributing to hospital admission cost [8, 9], AKI is independently associated with increased morbidity and mortality risks [10–12].

In this study, we sought to evaluate AKI risk associated with NSAIDs in hospitalized patients with different levels of baseline renal function. The study was facilitated by the richness of clinical data obtained from an electronic health record (EHR) architecture that had been previously established as part of a lager project, aimed at measuring and predicting risk for various preventable adverse drug events, including AKI [13].

Methods

Study population

This study included adult non-surgical patients (18 years or older) admitted to the University of Florida (UF) Health Shands Hospital between January 1, 2012, and September 30, 2015, or to the UF Health Jacksonville Hospital between March 1, 2013, and September 30, 2015. Both UF Health Shands and Jacksonville hospitals are public academic medical centers, housed within UF Health located in Gainesville, FL, USA. Patients with length of hospital stay less than 48 h or those with no recorded serum creatinine (SCr) value were excluded. Patients who were severely renal-compromised (i.e., those with end-stage renal disease (ESRD), renal cancer, congenital renal abnormality, and solid organ transplant) and those who had AKI or severe bleeding on admission were excluded [14, 15]. In addition, patients who received oral or parenteral ketorolac were excluded. Ketorolac, the most potent NSAID, has the unique property of posing the highest renal risk among NSAIDs and has a narrow set of indications, thus precluding its inclusion in an aggregate evaluation of NSAIDs [16]. The institutional review board (IRB) of the University of Florida approved this study.

Study design

Matching, index date, baseline period, and follow-up

We identified patients who had received at least one dose of NSAID during a hospital stay and defined them as exposed. The following oral or intravenous drugs were included in this study: aspirin (total daily dose > 400 mg), celecoxib, diflunisal, diclofenac, ibuprofen, etodolac, fenoprofen, indomethacin, ketoprofen, meclofenamate, meloxicam, mesalamine, naproxen, nabumetone, oxaprozin, piroxicam, sulindac, and salsalate.

The index date for the exposed group was the first day of use of any NSAID during the hospital stay. To establish a non-exposed group and to subsequently define their index dates, we selected one patient who had never received an NSAID during hospital stay per one exposed patient, matching on length of in-hospital stay and the presence of baseline renal impairment. Baseline renal impairment was defined by the presence of stage 3, 4, and 5 chronic kidney disease (CKD) using the *International Classification of Diseases, Ninth Revision, Clinical Modification* (ICD-9-CM) codes of 585.3-5 or 585.9 on the primary or secondary diagnoses at the current or previous admissions within the past year, or estimated baseline creatinine clearance (CrCl) using Cockcroft-Gault Equation of less than 60 mL/min. For the non-exposed group, the index date was the hospital day of the matching patient's index date, which is the day when their matching exposed patient received the first NSAID dose.

The baseline period included the index date and up to 2 days before the index date. Follow-up started from the day after index date. We excluded patients who developed AKI during the baseline period to minimize confounding by indication.

Stratification

We stratified the cohort by the presence of baseline renal impairment. Because ESRD (ICD-9-CM 585.6) was one of our exclusion criteria, patients who remained in the cohort and were considered to have baseline renal impairment are likely to have non-ESRD, but moderate-to-severe CKD. For the individual stratified cohorts and the pooled cohort, we used the same study design and analysis specifications as described in the following sections.

Exposure determination

Exposure was determined on a daily basis on index date and during follow-up period according to the actual exposure status. In other words, over time, patients could move between the exposed and non-exposed groups if they stopped or resumed NSAIDs.

Outcome determination

The primary outcome was (1) an increase in serum creatinine (SCr) to ≥ 150% within 7 days from baseline or (2) an increase in SCr by ≥ 0.3 mg/dL (26.5 μmol/L) within 48 h at any time during follow-up, according to the Kidney Disease Improving Global Outcomes (KDIGO) SCr-based criteria [17]. Baseline SCr was defined as the closest SCr to index date, which was measured during the baseline period. All pairs of SCr values (i.e., baseline SCr and all subsequent values within 7 days after the index date or any two SCr values measured after the index date and less than 48 h apart) were evaluated (Fig. 1). If any of the pairs met the outcome definition, the outcome date was the date of the second SCr value among the two SCrs compared. If multiple AKI events occurred, the earliest event was considered for the outcome.

Baseline covariates

Baseline covariates were chosen a priori to address clinically relevant confounders. First, patient demographics (i.e., age and gender) and results of different laboratory tests to define clinical risk factors including high body mass index (BMI), oliguria, hypovolemia, hypotension, anemia, albuminemia, elevated bilirubin level, and systemic infection were obtained from inpatients' and outpatients' EHRs. Second, laboratory tests are usually ordered in a certain clinical context and their absence is not random. Thus, the absence of monitoring of individual parameters, including albumin, urine output, blood urea nitrogen (BUN), SCr, and bilirubin, was also considered. We have reported the predictive properties of the absence of monitoring of AKI parameters elsewhere [18]. Patients' admission and discharge information, including date and time of admission and discharge, and diagnosis codes from billing data were also obtained. Diagnosis codes were utilized to define comorbid conditions, including CKD, hypertension, liver disease, diabetes, and cardiovascular disease (CVD). Physician orders and medication administration records during hospital stay were utilized to ascertain concurrent nephrotoxic medication and contrast agent use. Detailed definitions for baseline covariates are listed in Table 1.

A time-dependent covariate

NSAID use can be intermittent. Continuing, stopping, or resuming NSAIDs may affect AKI risk. To enhance NSAID exposure definitions, we introduced a time-dependent covariate that differentiated patients who discontinued NSAIDs from those who never received an NSAID (i.e., the non-exposure group) during their hospital stays and called them patients with "past NSAID use." To illustrate, past NSAID use is indicative of patient-days where exposed patients did not receive any NSAIDs on a particular day within 5 days after the first dose of NSAID during a hospital stay.

Analysis

We used time-dependent Cox proportional hazard models to compare AKI risk during patient-days on NSAIDs versus patient-days without NSAIDs. Censoring occurred either at 7 days from the index date or in the case of no administration of NSAIDs for 5 consecutive days, hospital discharge, or patient death, whichever came first. The Cox model included a binary exposure variable (a patient-day exposed to NSAIDs versus non-exposed) and was adjusted for past NSAID use as a time-dependent variable, in addition to above-mentioned baseline covariates. The development of an outcome was attributed to the exposure status on the preceding day.

Results

After application of the inclusion and exclusion criteria, our analysis included 55,808 of 123,995 admissions to the study sites during the study period (Fig. 2). Approximately 53% of the study population was female, with a mean age of 54.0

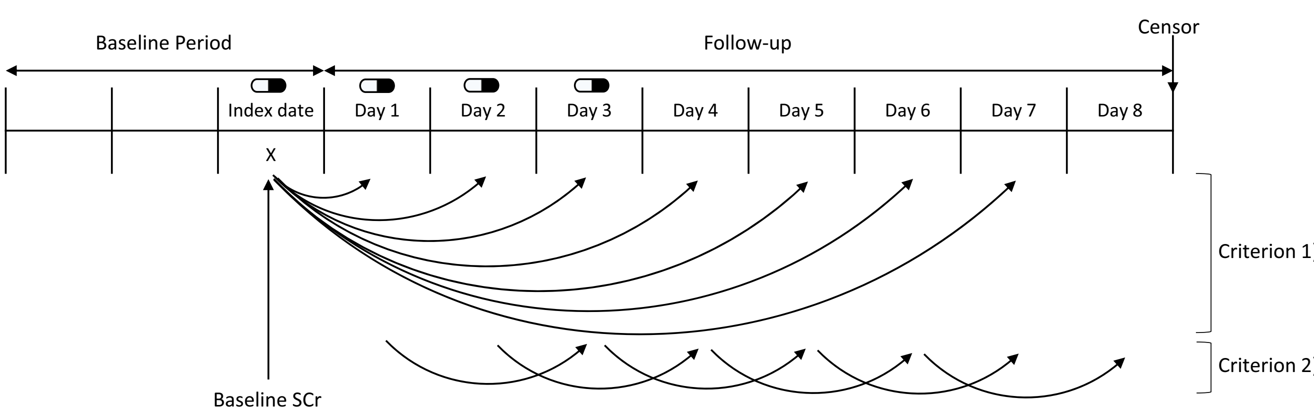

Fig. 1 Illustration of outcome ascertainment. Legend: Figure 1 illustrates outcome ascertainment in an example scenario where a patient received NSAIDs for 4 consecutive days and discontinued thereafter. By definition, the index date was the hospital day of first NSAID administration. Baseline periods include the index date and up to 2 days before the index date. Censoring occurred at 5 days after the last day of NSAID administration. All combinations of SCr measurements were compared and evaluated to see whether there was a significant increase in SCr; the significant increase in SCr is defined as (1) a SCr increase to ≥ 150% within 7 days or (2) a SCr increase by ≥ 0.3 mg/dL within 48 h at any time during follow-up

Table 1 Baseline covariates and definitions

Category	Data elements	Covariate name and definition	Look-back period†
Demographics	Gender	Gender	Admission day
	Birth date	Advanced age: age > 65	Admission day
	Body weight	Low body weight: body weight < 50 kg	Admission day
	Body weight and height	High body mass index: BMI > 35 kg/m^2	Admission day
	Location of ward	ICU: patient stayed in ICU	None
Laboratory values	Serum creatinine level, weight, gender	CrCl estimated by the Cockroft-Gault equation	30 days
	Serum albumin	Albuminemia: serum albumin < 3.0 mg/dL	None
	Urine output	Oliguria: urine output < 500 mL/day	None
	BUN and SCr	Dehydration: BUN/SCr ≥ 20	None
	Blood pressure	Low BP: SBP < 80 mmHg or MAP < 65 mmHg	None
	Body temperature, heart rate, respiratory rate, $PaCO_2$, and WBC	Systemic infection: presence of at least three of the following four clinical symptoms • Abnormal body temperature: < 36°C or > 38°C • High heart rate: > 90 beats/min • Abnormal respiratory function: RR > 20 or $PaCO_2$ < 32 mmHg • Abnormal WBCs: > 12,000/mm^3, < 4000/mm^3, or > 10% immature band	None
	Hgb, HCT	Anemia: Hgb <10 g/dL or HCT < 31%	None
	Serum bilirubin	Elevated bilirubin: serum bilirubin > 2.0 mg/dL	None
Absence of monitoring parameters	Albumin	Missing albumin value	None
	Bilirubin	Missing bilirubin value	None
	BUN and SCr	Missing BUN or SCr value	None
	Urine output	Missing urine output	None
Procedures	Physician orders	Procedures involving use of contrast	None
Comorbidities	ICD-9-CM codes	Chronic kidney disease	365 days
		Hypertension	365 days
		Liver injury	365 days
		Diabetes	365 days
		Heart failure	365 days
		Cardiovascular disease	365 days
		Coronary artery disease	365 days
Medications	Drug names and route	Use of high AKI risk medications, including amikacin, amphotericin B deoxylate, cidofovir, cisplatin, foscarnet, gentamicin, kanamycin, methotrexate, neomycin, polymyxin B sulfate, streptomycin, streptozocin, tacrolimus, tenofovir, or vancomycin	None
		Total number of AKI risk medications‡	None
		Use of ACEIs or ARBs	None
		Use of diuretics	None

†"Look-back" indicates the maximum duration prior to baseline for evaluating each data element utilized to define a variable

‡A list of AKI risk medications is available in Appendix A

Acronyms: *ICU*, intensive care unit; *CrCl*, creatinine clearance; *BUN*, blood urea nitrogen; *SCr*, serum creatinine; *BP*, blood pressure; *SBP*, systolic blood pressure; *DBP*, diastolic blood pressure; *MAP*, mean arterial pressure; *WBC*, white blood cell; *Hgb*, hemoglobin; *HCT*, hematocrit; *ICD-9-CM*, International Classification of Diseases, Ninth Revision, Clinical Modification; *AKI*, acute kidney injury; *ACEI*, angiotensin-converting enzyme inhibitor; *ARB*, angiotensin II receptor blocker

years (± 18.4) and median length of stay of 5 days (interquartile range [IQR]: 4–8 days). A total of 9860 admissions (17.7%) had at least one dose of any NSAID during their hospital stays. The median time to NSAID initiation from admission date was 1 day (IQR: 0–2). We observed that, of 29,743 hospital-days of exposure to different types of NSAID,

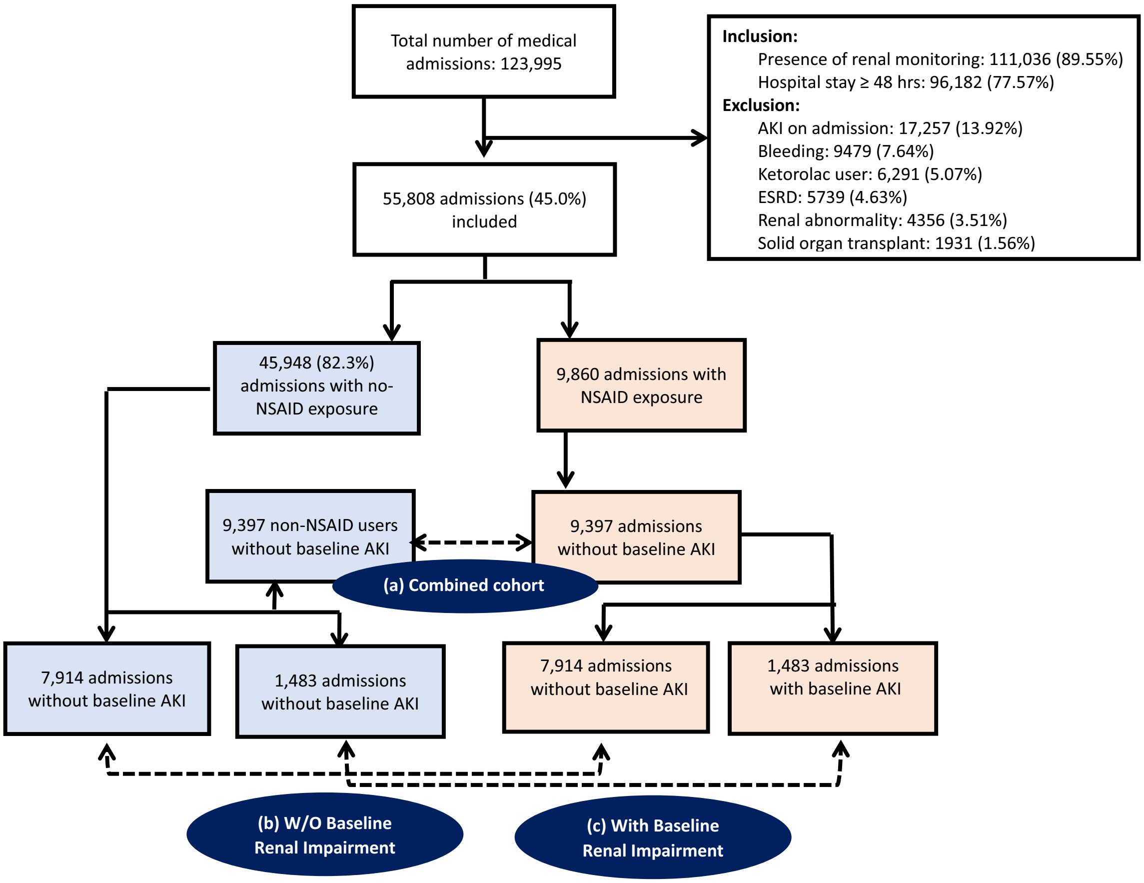

Fig. 2 Study cohort assembly flow chart

ibuprofen was predominantly used (19,051 hospital-days, 64.1%), followed by naproxen (2716 hospital-days, 9.1%) and celecoxib (2448 hospital-days, 8.2%).

As shown in Fig. 2, 18,794 admissions remained in the final combined cohort. Of 9397 admissions exposed to NSAIDS, 7914 admissions were in the without baseline renal impairment cohort (Fig. 2b) and 1483 admissions were in the with baseline renal impairment cohort (Fig. 2c) with the same number of matching non-exposed admissions in each of the stratified cohort.

Table 2 shows baseline characteristics by exposure in the pooled matched cohort. The non-exposed group was older and more likely to be intensive care unit (ICU) patients. In addition, a higher proportion of patients in the non-exposed group than in the NSAID user group had experienced comorbid acute/chronic conditions, such as albuminemia, elevated bilirubin, dehydration, CKD, liver injury, heart failure, diabetes, CVD, and CAD and received angiotensin-converting enzyme inhibitors (ACEIs), angiotensin II receptor blockers (ARBs), and diuretics. In addition, a higher proportion of patients in the exposed group than in the non-exposed group had experienced systemic infection and hypotension and received more nephrotoxic medications concomitantly with NSAIDs.

As expected, admissions with baseline renal impairment were substantially older (mean age: 68.1 vs 46.3 years) and stayed in the hospital longer (median: 6 vs 4 days) than those without renal impairment. Over 30% of patients also had diabetes, heart failure, and CAD, indicating that patients with renal impairment were at greater baseline risk for developing AKI. Baseline characteristics by exposure in each stratified cohort are provided in Appendix B. There are significant differences in baseline characteristics between the groups. In general, the exposed group had lower baseline AKI risk (i.e., less risk factors) than the non-exposed group, similar to what we observed in the pooled cohort. However, the exposed groups in the sub-cohorts and pooled cohort tend to receive more number of NTX drugs compared to non-exposed groups.

Table 3 shows AKI incidence by comparison groups within the pooled and stratified cohorts. The overall AKI incidence was 9.0 events/1000 patient-days. As expected, the incidence was more than three times higher in the "cohort with baseline renal impairment" than in the cohort without (23.3 vs 6.4 events/1000 patient-days; P-values < 0.001). The non-exposed groups, who had more risk factors for AKI, had slightly higher AKI incidences than the exposed groups in the pooled (9.5 vs 8.1 events/1000 patient days; *P*-

Table 2 Baseline characteristics of the exposure group compared with those in the total population

	Non-users *N*=9474	NSAID users *N*=9474	*P*-value
Demographics			
Age, mean (SD)	49.8 (16.3)	42.9 (16.3)	<0.001
Age > 65	2688 (28.6%)	1489 (15.7%)	<0.001
Gender, male	4675 (49.4%)	3439 (36.6%)	<0.001
Race			
White	5918 (63.1%)	5599 (59.7%)	<0.001
Black	2973 (31.7%)	3142 (33.5%)	0.0085
Others	488 (5.2%)	638 (6.8%)	<0.001
ICU stay	732 (7.8%)	375 (4.0%)	<0.001
Laboratory test-related			
Baseline CrCl, mean (SD)	80.0 (13.6)	82.1 (12.4)	<0.001
High BMI	694 (7.4%)	703 (7.5%)	0.8024
Albuminemia	788 (8.4%)	619 (6.6%)	<0.001
Elevated serum bilirubin	310 (3.3%)	234 (2.5%)	<0.001
Hypotension	797 (8.5%)	853 (9.1%)	0.1489
Dehydration	732 (7.8%)	488 (5.2%)	<0.001
Anemia	1904 (20.3%)	1801 (19.2%)	0.0590
Systemic infection	2317 (24.7%)	2645 (28.2%)	<0.001
Oliguria	1632 (17.4%)	1576 (16.8%)	0.2776
Absence of laboratory monitoring			
Albumin	4633 (49.4%)	4267 (45.5%)	<0.001
Bilirubin	4680 (49.9%)	4624 (49.3%)	0.4139
BUN to SCr ratio	2861 (30.5%)	2279 (24.3%)	<0.001
Urine output	3086 (32.9%)	3723 (39.7%)	<0.001
Medication-related			
Use of contrast	2326 (24.8%)	1941 (20.7%)	<0.001
Use of ACEIs and ARBs	2195 (23.4%)	1885 (20.1%)	<0.001
Use of diuretics	1388 (14.8%)	1022 (10.9%)	<0.001
Use of highly NTX drugs	1538 (16.4%)	1698 (18.1%)	<0.001
Total number of NTX drugs			
0	3648 (38.9%)	47 (0.5%)	<0.001
1	2645 (28.2%)	4202 (44.8%)	<0.001
2	1820 (19.4%)	2542 (27.1%)	<0.001
3	891 (9.5%)	1660 (17.7%)	<0.001
> 3	375 (4.0%)	929 (9.9%)	<0.001
Comorbid conditions			
Chronic kidney disease	816 (8.7%)	460 (4.9%)	<0.001
Hypertension	3958 (42.2%)	4230 (45.1%)	<0.001
Liver injury	591 (6.3%)	422 (4.5%)	<0.001
Diabetes	2514 (26.8%)	1876 (20.0%)	<0.001
Heart failure	1510 (16.1%)	1032 (11.0%)	<0.001
Cardiovascular disease	328 (3.5%)	131 (1.4%)	<0.001
Coronary artery disease	1801 (19.2%)	1397 (14.9%)	<0.001

Acronyms: *SD*, standard deviation; *ICU*, intensive care unit; *CrCl*, creatinine clearance; *BMI*, body mass index; *BUN*, blood urea nitrogen; *SCr*, serum creatinine level; *ACEIs*, angiotensin-converting enzyme inhibitors; *ARBs*, angiotensin II receptor blockers; *NTX*, nephrotoxic

Table 3 AKI risk associated with NSAID use

	Patient-days	Outcome	HRc†	HRa‡
Combined				
Non-exposed	39,377	375	Ref	Ref
Exposed				
NSAID use	23,785	192	0.80	1.01 (0.83–1.24)
Past NSAID use	14,949	138	0.92	0.97 (0.78–1.21)
Without baseline renal impairment				
Non-exposed	32,637	227	Ref	Ref
Exposed				
NSAID use	20,465	110	0.78	0.83 (0.63–1.08)
Past NSAID use	11,813	79	0.98	0.80 (0.60–1.07)
With baseline renal impairment				
Non-exposed	6443	145	Ref	Ref
Exposed				
NSAID use	3314	82	1.15	1.38 (1.04–1.83)
Past NSAID use	3122	59	0.92	0.94 (0.88–1.01)

†Adjusted by past NSAID use only

‡Adjusted by past NSAID use and baseline covariates, including patients' demographics, laboratory monitoring test results, absence of monitoring, nephrotoxic medication use, and comorbid conditions

Acronyms: *NSAIDs*, non-steroidal anti-inflammatory drugs; *HRc*, crude hazard ratio; *HRa*, adjusted hazard ratio

value=0.061) and without baseline renal impairment cohort (7.0 vs 5.4 events/1000 patient days; *P*-value=0.026), while the exposed group had a comparable AKI incidence in the "with baseline renal impairment" cohort. (22.5 vs 24.7/1000 patient days; *P*-value = 0.487).

After adjusting for potential baseline risk factors in addition to past NSAID use in a time-dependent manner, we found there was no significant association between NSAID use and hospital acquired AKI in the pooled cohort (HRa: 1.01 [95% CIs: 0.83–1.24]) and in the "without renal impairment cohort" (HRa: 0.83 [95% CIs: 0.63–1.08]). However, in the "with renal impairment cohort," the adjusted AKI risk was significant compared with the non-exposed group (HRa: 1.38 [95% CI: 1.04–1.83]).

Discussion

Our study assessed AKI risk following NSAID exposure in inpatient setting, adjusting for the aggregated effects of many other renal insults and susceptibilities. We also stratified the study cohort by baseline renal impairment to provide clinically meaningful information for improvement of NSAID use safety in hospitalized patients. This allowed the detection of elevated AKI risk when NSAIDs were used in patients with preexisting renal impairment at any doses, even for a single day exposure to an NSAID.

Among the approximately 3000 patients with pre-existing renal impairment included in our analysis, NSAID users were younger and had a smaller number of comorbid conditions than non-users. Even though avoidance of nephrotoxic medication is generally recommended for most CKD patients, we observed that they were more likely to receive several nephrotoxic medications concomitantly with NSAIDs. (Appendix B, Tables 1 and 2) This may reflect that patients with preexisting renal impairment present clinical challenges: they are susceptible to develop AKI, but pharmacological treatments are simultaneously necessary to manage their medical conditions, even if the treatments are nephrotoxic.

The prognosis of AKI among patients with CKD is known to be poorer than that of AKI in patients without preexisting renal complications. In a large observational study of patients with AKI requiring dialysis, more than 70% had baseline renal impairment [19]. Patients with renal impairment not only are vulnerable to AKI, which may or may not be associated with NSAID use, but also may be more likely to develop severe stages of AKI or other severe complications, have dialysis dependence, or die. Therefore, prevention of AKI in patients with preexisting renal impairment is particularly important. We suggest that the need for NSAIDs should be assessed prior to use and re-assessed with frequent renal function monitoring. Replacing NSAIDs with less nephrotoxic drugs or even avoidance of NSAID use may be acceptable in subsets of patients. As exposure to NSAIDs often cannot be avoided, gradual titration from low doses to the target doses, adjustment of NSAID dose based on accurate kidney function evaluation, and careful and adequate hydration to establish high urine flow rates should also be considered.

After adjustment for a large range of renal risk factors, our findings did not suggest that NSAIDs are independent contributors to increased risk of AKI in medical inpatients without renal impairment. Previous studies confirming NSAID-associated AKI risk were mostly conducted in outpatient settings [2, 20, 21]. The discrepancy between the results of the present study and those of previous studies may be explained by the differences in patient demographics especially age, duration of NSAID use, monitoring of renal function, and definition of AKI. Lafrance and Miller [21] found higher AKI risk in current (adjusted odds ratio (ORa): 1.82, [95% CI: 1.68–1.98]) and recent users (ORa = 1.60; 95% CI: 1.52–1.68) of NSAID in outpatient setting compared to non-users. Notably, almost half of the study population were over 65 years old and had chronic medical conditions. Huerta et al. and Griffin et al. evaluated the risk of hospitalization due to AKI associated with NSAID use in elderly and found current use of NSAIDs was associated with an increased risk of AKI. On the other hand, patients without renal impairment included in our study were relatively young with mean age of less than 50 and comorbid conditions were not as common as the patients included in the other studies.

Moreover, the median duration of NSAID use in our study population was 2 days, whereas the previous studies evaluated AKI risk in outpatient settings related to longer NSAID use. Lastly, AKI requiring hospitalization was the outcome of interest in the previous studies, whereas we assessed the mildest form of AKI (stage 1) as the primary outcome with a rationale that stage 1 AKI is an especially clinically relevant endpoint in the inpatient settings because of the associated immediate deleterious consequences, such as volume overload, electrolyte imbalances, metabolic acidosis, and uremia, which are known to increase inpatient mortality, hospital stay, and costs.

Because we used an aggregate NSAID exposure definition, our results were largely driven by the most prevalent NSAID types, mainly ibuprofen. A recent inpatient setting study evaluated the risk of 29 nephrotoxic medications simultaneously after controlling for other underlying differences in AKI risk, and found that ibuprofen, the only NSAID included in the analysis, was not associated with increased renal risk [22], which is consistent with our results. We did not perform an analysis of individual NSAIDs owing to limited utilization of each NSAID in our study cohort. Thus, the heterogeneity of AKI risk among different NSAIDs needs to be investigated in future studies with adequate sample size for subtype analyses.

This study presents several strengths from analytical and clinical perspectives. First, we operationalized and incorporated a wide range of clinically important risk factors obtained from discrete fields in hospitals' EHR system. This allowed an assessment of NSAID-associated AKI risk while simultaneously accounting for underlying susceptibilities and inpatient care complexities, such as inadequate monitoring of several clinical patient parameters. Second, the baseline SCr was defined from the immediate time period before the initiation of a NSAID, which facilitated the detection of AKI rates associated with NSAID use in line with Kidney Disease Improving Global Outcomes (KDIGO) definitions. Traditionally, studies on drug-induced nephrotoxicity in hospitals have relied on manual extraction of information from charts, which has often resulted in the inclusion of a limited number of confounders in analysis or an overly simplified determination of baseline SCr [23]. For example, studies have defined baseline SCr using the values measured upon hospital admission [24–26] or did not specify their baseline SCr definition [27, 28]. Defining baseline SCr with the value obtained on admission cannot control other renal insults that occur between admission and treatment initiation, and may result in outcome misclassification. Finally, residual AKI risk was evaluated up to 5 days after withdrawal of NSAIDs by employing a time-dependent variable (i.e., past NSAID use) that indicated patient days after discontinuation of NSAIDs.

We observed a few limitations of our study. First, limited sample size did not allow exploration of how long a certain level of risk remained after an NSAID dose. We hypothesized that NSAID-associated AKI may manifest within up to 5 days of NSAID discontinuation, considering a possible delayed progression of renal function decline. Because renal elimination of NSAIDs varies substantially across patients, the 5-day exposure time window is subject to variation and may erroneously attribute time to no NSAID exposure despite potential residual NSAID risk on the kidneys. To test this, a larger study in the future might consider a continuous variable measuring cumulative days after discontinuation of NSAIDs, and gauge appropriate risk windows for follow-up.

Although SCr is one of the most common routine laboratory monitoring parameters in hospitals, it is not monitored daily in routine clinical care. If detection of NSAID-associated AKI was delayed owing to sporadic SCr monitoring, outcomes could be misattributed to non-exposure time. To address outcome misclassification due to missing SCr, multiple imputations for missing SCr values may be an option in future analysis, as they have shown improvement of specificity and positive predictive value for detecting AKI [29, 30].

Second, we did not assess renal risk factors that may have emerged after NSAID initiation with an assumption of no time-varying confounding. We did so because a subtle increase in SCr before a 0.3 mg/dL (26.5 μmol/L) or 50% increase (AKI) may not be associated with NSAID discontinuation because it does not pose a significant clinical concern. Future studies may use an advanced analytical method, such as marginal structural model (MSM), to test for possible time-varying confounding.

Lastly, among 18,548 censored patients, 17,618 patients (95.0%) were censored at hospital discharge. Since we did not have follow-up data beyond discharge, we assumed that these patients would have been discharged because they were healthy enough not to develop AKI after leaving the hospitals. Even if AKI events occurred after discharge, we assumed that the relative risk of undetected AKI events would be likely similar with that in patients who remained on the study. However, one other censoring criterion, which was differentially applied according to exposed and non-exposed time, was termination of follow-up at 5 days after NSAID discontinuation. This may potentially be the source of informative censoring [31]. The censoring criterion only affected non-exposed time among NSAID users; thus, this resulted in the absence of follow-up time for patients who were non-exposed at the time of censoring. This would have led to an overestimated AKI risk of exposed time to NSAIDs. Under this circumstance, the association would have been protective in an unbiased setting, which may hardly be explained by NSAID mechanism and AKI etiology. Future studies may need to consider advanced statistical methods (e.g., inverse probability censoring weighting) to address possible bias due to informative censoring.

Conclusion

Our study demonstrated that short-term NSAID use is associated with an increased risk for AKI in hospitalized patients with preexisting renal impairment. Given that the prognosis of acute on chronic kidney failure is known to be poorer, the renal risks of NSAID use may outweigh its benefits. On the other hand, we found no elevated risk in hospitalized patients without baseline renal impairment, who were relatively healthier and younger than the patients with preexisting renal impairment.

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Author contribution Conceived and designed the analysis: NJ, RS, BB, AW

Obtained the data: NJ

Contributed data or analysis tools: NJ

Performed the analysis: NJ

Wrote the paper: NJ, RS, BB, AW

Declarations

Conflict of interest The authors declare no competing interests.

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